The Bombay Salesian Society's
DON BOSCO INSTITUTE OF TECHNOLOGY
An Autonomous Institute Affiliated to University of Mumbai
Premier Automobiles Road, Kurla West, Mumbai - 400070

Report on - Arduino - Hands-on Workshop

Title: Arduino - Hands-on Workshop
Date: 9th December, 2025
Time: 9:30 am - 11:30 am
Venue: First Floor, C-Wing, RF- Lab

Target Audience: SE IEEE Members

No. of Participants Present: 4

No. of Girl Participants Present: 2

No. of Boy Participants Present: 2

Organizing Department / Committee / Authority: IEEE - RAS

Faculty Coordinator: Mrs. Freda Carvalho

Objectives:

❖To analyze the architecture of the Arduino board, including microcontroller functionality, GPIO (digital/analog pins), and power management.

❖To understand embedded C-based programming in the Arduino IDE, including pin configuration, control logic, and serial communication protocols.

❖To implement real-time interfacing techniques using sensors and actuators, including PWM signal generation and ultrasonic signal processing.

Outcomes:

❖Acquired technical proficiency in configuring GPIO pins, generating PWM signals, and handling analog/digital I/O operations

❖Successfully developed and tested embedded applications such as LED control using digital output, brightness modulation via PWM, and distance measurement using an ultrasonic sensor.

❖Gained practical exposure to debugging using serial communication and improved understanding of real-time embedded system design and implementation.

Detailed Report:

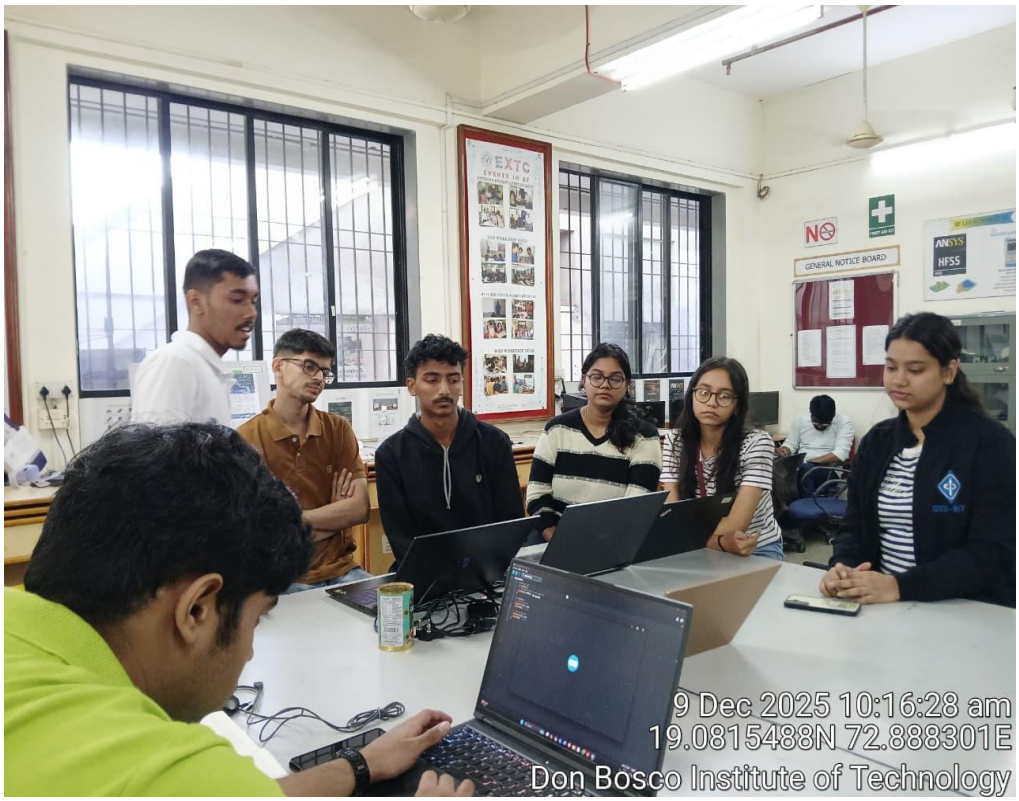
An Arduino Hands-on Workshop was conducted to introduce participants to the fundamentals of Arduino hardware and programming. The session provided comprehensive knowledge of the Arduino board structure, including the microcontroller, digital and analog pins, power pins, and communication interfaces such as USB and power jack. The functionality of each component and their role in input-output operations with sensors and actuators were clearly explained.

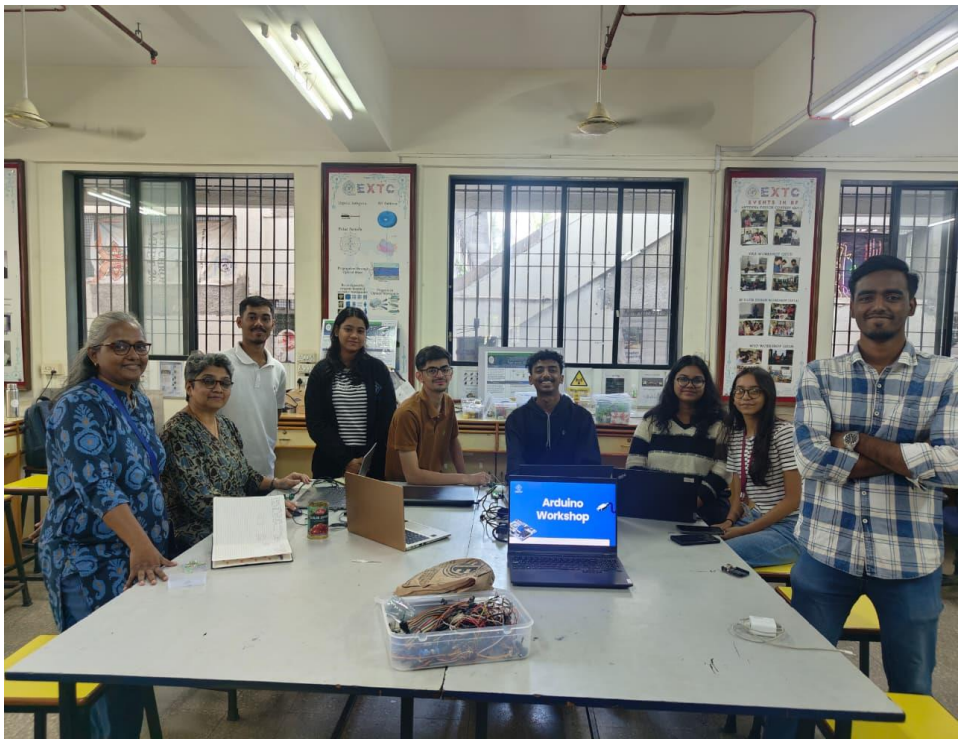
Participants were introduced to the Arduino IDE and basic programming structure, focusing on the two essential functions: setup(), used for initialization, and loop(), which executes the main logic continuously. The process of code uploading, serial communication for debugging, and interfacing external components was also demonstrated.

During the practical session, three mini-projects were completed: LED ON/OFF control using digital pins, LED intensity variation using PWM, and distance measurement using an ultrasonic sensor up to 50 cm.

The workshop proved to be highly informative, enhancing understanding of embedded systems and providing valuable hands-on experience in Arduino-based application development.

Snapshot of the Event:





Social Media Links:

https://www.instagram.com/ieee_dbit/

Details of the attendees:

S. No.	Name	Branch
1	Musham Hasan	EXTC
2	Neha Haldankar	EXTC
3	Prachi Mandlik	EXTC
4	Vineet Suryawanshi	EXTC

Report Prepared By: IEEE-DBIT

Name of the Student: Aarya Jadhav & Shreya Manoj

Post of the student:
Reporting Head & Joint Reporting Head, IEEE-DBIT

Report Approved By: IEEE - DBIT

Name of the Faculty: Ms. Freda Carvalho

Post of the Faculty: IEEE-DBIT Student Branch Counsellor